

Native Thermodynamic Curvature

Ordered response, Smriti balance, and the Hessian representation

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Result and source contract

The thermodynamic response paper already contains exact potential differentials, six unit-typed responses, a covariant response tower, and loop recognition with a prior ledger. The later TC and PF chapters construct Hessian response geometry. This edition supplies their missing **explicit EMK connection intertwiner**, then extends the scalar ledger to ordered native transport. It consumes the existing algebra and calculus; it does not propose a second operator engine.

The principal native equations are

$$\begin{aligned}\mathbf{H} &= \frac{a+c}{2}1 + \frac{a-c}{2}K + bRK, & X_i &= \mathbf{H}^{-1}\delta_i\mathbf{H}, \\ \mathcal{F}_{ij}^H &= -\frac{1}{4}[X_i, X_j], \\ \mathcal{F}^{\text{open}} &= \mathcal{F}^A - D_A M + M \wedge M.\end{aligned}$$

Here the first curvature is selected response transport; the second compares a connection with a **previously declared**, same-carrier Smriti ledger. They are not assumed to coincide with every use of the word curvature in the archive.

The sources are Publications main 121f2dff607fbe0a8b8550fa19b97637ed82509b (EMK-C1), its research edition f0f1c1650e914b7eaf3bf64ddb7df9f543c92a56 (corrected thermo manuscript, TC, PF and CP), and canonical RKF main 3cc5a33b05c16d59c90994ddda69dedc0d392424. SOURCE_MAP.md identifies the exact chapters and all thermodynamic theorem capsules. SOURCE_PINS.json binds their bytes. The printed equations below have written proofs; the certificate checks a declared finite rational sector.

The source order is: native algebra and lawful derivatives; response target and constitutive chart; connection selection; curvature and memory; observer; then the faithful tangent representation. An equation of state and entropy/volume calibration remain physical input. No universal material, clock or spacetime metric is inferred from an abstract algebra relation.

NT-1. Native response carrier and the six thermodynamic coordinates

Work over the radial real sector of the established cut scalars, extended by positive square roots when used below. The algebra is the existing associative EMK algebra

$$K^2 = 1, \quad R^2 = -1, \quad KR = -RK, \quad K^\dagger = K, \quad R^\dagger = -R.$$

Put $L = RK$. Then $L^\dagger = L$, $L^2 = 1$, and

$$[R, K] = 2L, \quad [K, L] = -2R, \quad [R, L] = -2K.$$

The four native normal words are $1, K, R, L$. The internal element R is not silently identified with the central cut scalar ι_Σ .

Let sc select the coefficient of 1. Native multiplication gives

$$\text{sc}(xy) = \text{sc}(yx), \quad \text{sc}(x^\dagger x) = x_0^2 + x_K^2 + x_R^2 + x_L^2.$$

For radial functions a, b, c with $a > 0$ and $\Delta = ac - b^2 > 0$, define $\mathbf{H}$ as above. Then

$$\mathbf{H}^{-1} = \frac{(a+c)1/2 - (a-c)K/2 - bL}{\Delta}.$$

In particular, this is a native inverse, before a matrix interpretation.

Proof. The mixed terms in the square of $(a-c)K/2 + bL$ cancel because $KL = -LK$. Its square is $((a-c)^2/4 + b^2)1$. Multiplication by the proposed numerator leaves $\Delta 1$ on both sides. The scalar-part identities follow from the same four-word table.

There is also a native positive factorization. Set

$$P_\pm = (1 \pm K)/2, \quad E_{12} = (L - R)/2, \quad B = \sqrt{a} P_+ + \frac{b}{\sqrt{a}} E_{12} + \frac{\sqrt{\Delta}}{\sqrt{a}} P_-.$$

The relations give $B^\dagger B = \mathbf{H}$, and B is invertible. Thus $\text{sc}(x^\dagger \mathbf{H} x) = \text{sc}((Bx)^\dagger Bx) > 0$ for $x \neq 0$. This constructs a positive finite native pairing from the declared response element; an ordinary Hilbert space was not its primitive input.

For the thermodynamic adapter, choose constant positive units and translated dimensionless coordinates s, v , a supplied C^4 energy $u(s, v)$, and the stable chart

$$a = u_{ss}, \quad b = u_{sv}, \quad c = u_{vv}, \quad \Delta > 0.$$

The physical identification is the one in the source paper. In dimensional coordinates temporarily use $a = U_{SS}, b = U_{SV}, c = U_{VV}$. Differentiating $T = U_S, P = -U_V$, with the source's fixed-amount convention, gives

$$\begin{aligned} dU &= T dS - P dV, & T_V &= -P_S, \\ C_V &= T/a, & C_P &= Tc/\Delta, & K_S &= Vc, & K_T &= V\Delta/a, \\ \lambda_P &= -S\Delta/c, & \lambda_V &= -Sa, & \lambda_S &= -Vc, & z_T &= -V\Delta/a, \\ z_S &= P/c, & \lambda_T &= Pa/\Delta. \end{aligned}$$

To prove these formulas, impose respectively $dV = 0, dP = 0, dS = 0, dT = 0$ in $dT = a dS + b dV$, $dP = -b dS - c dV$; substitute the reciprocal derivatives in the six source definitions. The four Legendre differentials follow by differentiating $U + PV, U - TS, U + PV - TS$. Consequently $\Gamma_c \Gamma_m = (C_V/C_P)(K_S/K_T) = 1$.

The six responses retain their temperature, pressure and volume units until nondimensionalized. The older *uniform entropy-scaled tetrad* $\lambda_X^{\text{th}} = -ST/C_X$ is a different declared coordinate family. In

particular its two mechanical coordinates must not replace λ_S, z_T in the six-response formulas. Scalar ratio closure is not full curvature flatness, as NT-4 will demonstrate on this very same stable chart.

NT-2. Curvature is the native transport-square defect

Use the EMK-C1 derivative contract: derivations δ_i satisfy $[\delta_i, \delta_j] = c_{ij}^k \delta_k$, with central constant frame coefficients obeying Jacobi. On the regular right module define $\nabla_i x = \delta_i x + A_i x$. The curvature coefficient is

$$F_{ij}^A = \delta_i A_j - \delta_j A_i + [A_i, A_j] - c_{ij}^k A_k.$$

It satisfies

$$([\nabla_i, \nabla_j] - c_{ij}^k \nabla_k)x = F_{ij}^A x.$$

For a two-sided invertible native change g ,

$$A'_i = g^{-1} A_i g + g^{-1} \delta_i g, \quad F_{ij}^{A'} = g^{-1} F_{ij}^A g.$$

Proof. Expand the two compositions with Leibniz; the terms differentiating x cancel after subtracting the frame bracket. Conjugate ∇_i by left multiplication with g to obtain both transformation identities. This is EMK-C1 specialized to the thermodynamic coefficient carrier.

Hence neither ordinary off-diagonal entries nor an uncorrected frame commutator define this curvature. For example $[R, L] = -2K$ is nonzero and diagonal in the later standard cut chart. Conversely a noncommuting direction frame with $A = 0$ has $F^A = 0$ relative to that frame. Even central coefficients can have curvature through their derivative terms.

For coordinate directions, parallel transport is fixed here by $\dot{w} = -A(\dot{\gamma})w$, with later segments multiplying on the left. A positive rectangle follows $+i, +j, -i, -j$; its holonomy is $1 - \varepsilon^2 F_{ij} + o(\varepsilon^2)$. Reversing the rectangle reverses this sign. For constant coefficients this follows by multiplying $e^{\varepsilon A_j} e^{\varepsilon A_i} e^{-\varepsilon A_j} e^{-\varepsilon A_i}$; allowing variable coefficients adds the two derivative terms with the same sign. This explicitly fixes the orientation that an unnamed “small square” would leave ambiguous.

NT-3. Select response transport and derive its native curvature

Assume now that δ_i commutes with dagger. A compatible connection for $h_H(x, y) = x^\dagger \mathbf{H} y$ obeys

$$\delta_i \mathbf{H} = A_i^\dagger \mathbf{H} + \mathbf{H} A_i.$$

Select the strain representative by the additional condition $A_i^\dagger \mathbf{H} = \mathbf{H} A_i$. Then it is unique and equals

$$A_i^H = \frac{1}{2} X_i, \quad X_i = \mathbf{H}^{-1} \delta_i \mathbf{H}.$$

Without the additional condition there is the genuine freedom $A_i = A_i^H + Q_i$, $Q_i^\dagger \mathbf{H} + \mathbf{H} Q_i = 0$. The selection is made in the declared response trivialization, then transported as a connection; it is not reimposed after every variable frame change.

The native inverse derivative and derivative-frame relation give

$$\delta_i X_j - \delta_j X_i - c_{ij}^k X_k = -[X_i, X_j].$$

Consequently for every central constant q ,

$$F_{ij}^{qX} = q(q-1)[X_i, X_j], \quad \boxed{F_{ij}^H = -\frac{1}{4}[X_i, X_j]}.$$

Proof. The two selection equations imply $2\mathbf{H}A_i = \delta_i \mathbf{H}$, proving uniqueness. Conversely this choice satisfies both equations because $(\delta_i \mathbf{H})^\dagger = \delta_i \mathbf{H}$. Differentiating $\mathbf{H}^{-1} \mathbf{H} = 1$ gives $\delta_i \mathbf{H}^{-1} = -\mathbf{H}^{-1}(\delta_i \mathbf{H})\mathbf{H}^{-1}$. Apply it to X_j , antisymmetrize, and subtract the frame term. The second derivatives cancel by the derivative contract. Substitute into NT-2.

The coefficient $q = 1$ is pure gauge and flat, even when the X_i do not commute. The metric-compatible strain choice is $q = 1/2$, not $q = 1$, wherever $\delta \mathbf{H} \neq 0$. The minus sign and quarter factor therefore carry real content. Response rotation alone does not imply curvature; the obstruction is this selected transport commutator.

NT-4. Exact classical special case and an oriented native marker

Let $e = P_+$. The native left ideal $\mathcal{E} = \mathcal{A}e$ is a rank-two module over the central radial functions, with basis e, Re . Left multiplication is faithful and gives

$$\rho(K) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \rho(R) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \rho(L) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \rho(\mathbf{H}) = \begin{pmatrix} a & b \\ b & c \end{pmatrix} = H.$$

This module is not asserted to be a right module over all of $\mathcal{A}$. Its right scalars are the central coefficient functions. The pairing $2\text{sc}(x^\dagger \mathbf{H}y)$ on $\mathcal{E}$ is exactly the response form with matrix H .

For commuting coordinate derivatives and $H = D^2u$, identify e, Re with the normalized s, v tangent directions. Then

$$\rho(A_i^H) = \frac{1}{2}H^{-1}\partial_i H = \Gamma_i, \quad \boxed{\rho(F_{sv}^H) = R_{sv}^g}.$$

Here $g = H_{ij}dx^i dx^j$ and R^g is its Riemann curvature in the convention $[\nabla_s, \nabla_v]$. This closes the *specific* EMK-to-Hessian connection map left open in TC-5.

Proof. The matrices follow by acting on e, Re . They form a basis of all real two-by-two matrices, proving faithfulness; multiplication and dagger are respected. Third derivatives of u are completely symmetric, so $\Gamma_{ij}^k = \frac{1}{2}H^{k\ell}u_{\ell ij}$ is torsion free. NT-3 supplies metric compatibility, which makes this the Levi-Civita connection. The representation respects derivatives and products, so it intertwines connections and curvature. No identification of a general native connection with a tangent connection has been assumed.

Define native $J_H = R\mathbf{H}/\sqrt{\Delta}$. Then

$$J_H^2 = -1, \quad J_H^\dagger \mathbf{H} = -\mathbf{H}J_H, \quad \partial_i J_H + [A_i^H, J_H] = 0.$$

The first two follow by multiplication; differentiation of the first expression using NT-3 proves the last (equivalently expand its four native coefficients). Hence $[F^H, J_H] = 0$. Metric compatibility also makes F^H skew for the H -pairing. In this two-dimensional response sector the intersection is the line spanned by J_H , so

$$F_{sv}^H = J_H f_{sv} = -\kappa R \mathbf{H}, \quad f_{sv} = -\kappa \sqrt{\Delta}.$$

In native coefficients the signed readouts are

$$\boxed{f_{sv} = -\text{sc}(J_H F_{sv}^H), \quad \kappa = \frac{\text{sc}(R \mathbf{H} F_{sv}^H)}{\Delta}.}$$

The scalar part of F^H itself is zero. Thus an ordinary trace misses nonzero curvature, while this calibrated marker recovers its orientation. Both the marker and curvature must be transported under a change of response frame. In the tangent adapter $\kappa = (H R_{sv}^g)_{12}/\Delta$ is Gaussian curvature. Outside that adapter it is the stated response-coordinate marker.

Conjugating by the native factor B of NT-1 gives

$$B A^H B^{-1} - d B B^{-1} = R \alpha, \quad \alpha = \frac{db - (b/a) da}{2\sqrt{\Delta}}, \quad d\alpha = f_{sv} ds \wedge dv.$$

Indeed the transformed compatible connection is skew, hence a multiple of R ; its lower-left coefficient gives the displayed α . This is the exact PF-1 phase sector within the native connection.

For the source fixture

$$u = 10s - 10v + s^2 + sv + \frac{5}{2}v^2 + \frac{1}{2}s^2v$$

at the origin, $(a, b, c) = (2, 1, 5)$, $\Delta = 9$. Direct native multiplication gives

$$F_{sv}^H = \frac{5}{324}K - \frac{35}{648}R + \frac{5}{216}L, \quad \kappa = \frac{5}{324}, \quad f_{sv} = -\frac{5}{108}.$$

All caloric-mechanical ratio identities hold here while this curvature is nonzero. Deleting only the cubic term leaves the same point Hessian and all six point responses, but makes the curvature zero. Thus the higher response tower is necessary; a point spectrum or scalar closure ratio does not determine it. Positive physical entropy and volume origins can be translated as in TC-5.

NT-5. Non-Abelian Smriti balance and its scalar seam sector

Let A be a connection and let M be a prior native endomorphism-valued one-form on the **same** carrier. Under frame change it transforms tensorially, $M' = g^{-1} M g$. Define the open connection $C = A - M$. Put

$$(D_A M)_{ij} = \delta_i M_j - \delta_j M_i - c_{ij}^k M_k + [A_i, M_j] - [A_j, M_i].$$

Then

$$\boxed{F_{ij}^C = F_{ij}^A - (D_A M)_{ij} + [M_i, M_j].}$$

In differential-form notation $(M \wedge M)_{ij} = [M_i, M_j]$. Thus native local balance is the exact equation

$$\boxed{F^A = D_A M - M \wedge M,}$$

which permits $F^A \neq 0$. The equation is gauge covariant.

Proof. Expand $F(A - M)$ in NT-2 and collect derivative, cross and quadratic terms. Since M transforms tensorially, C transforms as a connection and its curvature transforms by conjugation. This proves the balance equation and its covariance. It is generally false that $F(A - M) = F(A) - F(M)$.

For constant $A_s = K, A_v = L$ and $M_s = 0, M_v = L$, $F^A = -2R$, $D_A M = -2R$, $M \wedge M = 0$, and $F^C = 0$. The ledger is declared as part of this model before testing the loop. An arbitrary post-hoc choice $M = A$ is not evidence that a physical memory law has been derived.

The original paper's additive response law is an exact special sector. Take a fixed nonzero native generator Z and radial one-forms ω, ℓ , with $A = Z\omega, M = Z\ell$. All coefficient commutators vanish, giving

$$F^C = Z d(\omega - \ell), \quad \oint_{\partial S} (\omega - \ell) = \int_S d(\omega - \ell).$$

For the thermo phase line $Z = R$, NT-4 supplies $\omega = \alpha$. On a simply connected smooth domain, $d(\omega - \ell) = 0$ is equivalent to every additive loop residue vanishing, by path integration. On a domain with holes the periods remain data. Principal phase also loses full turns; the lifted integral is retained as Smriti even when its exponential is one.

If $\eta = \omega - \ell$ is piecewise smooth across a transverse seam $r = 0$, the source SC-03/SC-05 identity specializes to

$$d\eta = (1 - H(r))d\eta_- + H(r)d\eta_+ + \delta(r) dr \wedge (\eta_+ - \eta_-).$$

It follows by differentiating $\eta_- + H(r)(\eta_+ - \eta_-)$. For $\eta_- = 0, \eta_+ = c dv$, a positive rectangle crossing the seam with height b has loop residue cb , despite flat smooth pieces. The seam and bulk channels cannot be silently cancelled across different supports. For general noncommuting coefficients, ordered transport replaces the scalar Stokes readout; no ordinary exponential of a surface integral is asserted here. History-dependent memory requires its declared extended state.

NT-6. What an observer retains: descent versus compression

For a surjective bundle readout $O : \mathcal{E} \rightarrow \mathcal{Y}$, the connection descends through O exactly when its kernel subbundle is preserved by ∇ . In that case

$$O\nabla_i = \bar{\nabla}_i O, \quad OF_{ij} = \bar{F}_{ij} O.$$

For each curvature target, faithful flatness additionally requires $\ker O \cap \text{ran } F_{ij} = 0$. Descent alone is insufficient.

Proof. Define $\bar{\nabla}_i(Ox) = O\nabla_i x$. Independence of the representative is exactly kernel preservation. Composing and subtracting the frame term proves curvature descent. The displayed intersection condition is precisely injectivity of O on the curvature output range.

Compression is a different operation. For a smooth native idempotent P , set $Q = 1 - P$ and $D_i P = \delta_i P + [A_i, P]$. The connection $\nabla_i^P = P\nabla_i$ on $\text{ran } P$ has curvature

$$F_{ij}^P = PF_{ij}^A P + P[D_i P, D_j P]P.$$

For a constant cut,

$$F_{ij}^P = PF_{ij}^A P - PA_i QA_j P + PA_j QA_i P.$$

To prove the formula, expand $P\nabla_i(P\nabla_j x)$, use $Px = x$, $P(D_i P)P = 0$, and $Q\nabla_i x = (D_i P)x$. Antisymmetrization gives the displayed correction. It is the same hidden-return channel as the native identity $PabP - PaPbP = PaQbP$, now with both transport orders retained.

For $A_s = R, A_v = L, P = P_+$, both compressed coefficients vanish, so $F^P = 0$. Nevertheless

$$F^A = -2K, \quad PF^A P = -2P, \quad PA_s QA_v P - PA_v QA_s P = -2P.$$

The complement return cancels in the compressed verdict, without erasing the full native curvature. The actual quotient descent test fails because $PA_s Q \neq 0$. This is a constructive example of observer-dependent flatness, not a theorem that all observation flattens curvature.

NT-7. The precise Onsager-flatness statement

In a calibrated real force chart $x = (x^1, \dots, x^m)$, suppose the flux law is $j = L(x)x$, with the conjugate pairing used for entropy production. Use the native central connection $A = 1\beta$, $\beta = \sum_i j_i dx^i$. Then

$$F_{ij} = 1 \left[L_{ji} - L_{ij} + \sum_k (\partial_i L_{jk} - \partial_j L_{ik}) x^k \right].$$

For **constant** L , flatness is equivalent to $L = L^T$, and

$$F_{ij} = -2(L_a)_{ij}1, \quad \sigma = x^T L x = x^T L_s x, \quad L_s = (L + L^T)/2, \quad L_a = (L - L^T)/2.$$

Thus constant reciprocal transport is a flat *process-form sector*. This is not a flatness criterion for the different response-metric connection in NT-3.

Proof. Differentiate $j_j = \sum_k L_{jk} x^k$ and antisymmetrize. The native coefficient commutator is zero in this central sector, but the covariant-derivative commutator need not be. The scalar $x^T L_a x$ equals its negative after transposition and hence vanishes.

For variable L , pointwise reciprocity leaves the derivative term. An exact positive counterexample is

$$L(x) = \text{diag}(1, 1 + x^1), \quad x^1 > -1.$$

It is symmetric and positive, but $F_{12} = x^2 1$. On the other hand constant $L = I$ has positive production away from zero and flat β , while constant $L = \rho(R)$ has zero production and nonzero curvature. There is therefore no universal equality of entropy production and curvature.

Under a constant dual calibration $j' = B j, x' = B^{-T} x$, both $j'^T x' = j^T x$ and $j'^T dx' = j^T dx$ are invariant, with $L' = B L B^T$. This recovers the original paper's reciprocal covariance. Time-reversal parities, kinetic coefficients and the physical identification of σ remain those of the declared Onsager adapter.

NT-8. Native response tower and the surviving theorem ledger

Let E be the unit-typed six-response module after lawful normalization, with a declared fibre connection, and let the base cotangent connection be the one selected in NT-3/NT-4 when that adapter is used.

The source tower is

$$L_1 = \hat{\Lambda}, \quad L_{n+1} = \nabla L_n, \quad L_n \in E \otimes (T^*)^{\otimes(n-1)}.$$

Its curvature is forced by the product rule:

$$(\mathcal{F}_{ij}^{(n)} T)(v_1, \dots, v_{n-1}) = F_{ij}^E T(v_1, \dots, v_{n-1}) - \sum_r T(v_1, \dots, F_{ij}^H v_r, \dots, v_{n-1}).$$

This supplies the native curvature action on the response theorem's covariant tower. It does not confuse the six-dimensional response fibre with the two-dimensional tangent module. A flat internal connection on the six scaled scalar components is one choice, not a consequence of the Hessian formula.

Proof. Apply the connection product rule twice to evaluation of a tensor. The mixed first-derivative terms cancel in the commutator; the derivative-frame terms subtract on every slot. A covector acts with the negative transpose action because $\delta\langle\xi, v\rangle$ is the sum of the two covariant derivatives. Summing slots proves the formula. It is unchanged under a faithful intertwiner such as NT-4.

The algebraic Cut-Flow-Jet results require only their original native module and composition contracts. For a finite raiser $\mathbb{R}^{N+1} = 0$, $\text{Exp}(t\mathbb{R}) = \sum_{k=0}^N t^k \mathbb{R}^k / k!$. If $[\mathbb{T}, \mathbb{R}] = 0$, the ordered Cauchy product gives $\text{Exp}(t(\mathbb{T} + \mathbb{R})) = \text{Exp}(t\mathbb{T}) \text{Exp}(t\mathbb{R})$ as formal series (and in the source's admitted convergent completion). Holonomic identification with $\text{Sym } \nabla^n \Lambda$ is still required. For an involution J , define $G_e = (G + JGJ)/2$, $G_o = (G - JGJ)/2$. Direct multiplication through order two followed by $\log(1+Y) = Y - Y^2/2 + \dots$ yields

$$\log(J \text{Exp}(tG) J \text{Exp}(tG)) = 2tG_e + t^2[G_e, G_o] + O(t^3).$$

That coefficient is the original cut-loop target. It equals a constant two-direction connection curvature when $A_1 = G_e$, $A_2 = G_o$; it is not automatically F^H . Adjoint orthogonality and the clock-free Eye theorem retain their extra positive/spectral hypotheses. Changing their symbols to native words does not establish those hypotheses anew.

Certification and what this conversion establishes

The conversion constructs an exact native response element, selects its connection under a stated condition, and proves a faithful thermo tangent intertwiner. It also proves the non-Abelian ledger balance, compression residue, Onsager process-form criterion and induced tower action. These are mathematical bridges within the established programme, with standard connection and Hessian identities explicitly attributed through the source map. External priority is not claimed.

The executable package uses the unchanged canonical Pāṇinian compiler for native words. An independent Python Fraction implementation differentiates the full Christoffel formula, including fourth-order jets, to check the tangent adapter. Negative controls distinguish closure ratios, traces, derivative terms, ledger cross terms, hidden returns and orientation signs. The certificate enumerates its actual executed scope; source pins alone do not recertify all old capsules. Proof-assistant formalization and independent external review are not claimed.

The connection selector, material energy law, units, preparation and physical observer remain declared inputs. In particular this paper does not infer spacetime curvature, a universal dissipation law, an arithmetic source identity, or new experimental validation. The original thermo paper and source archives remain preserved. This native edition is an additive conversion and explicit bridge, not a change to the already released arXiv record.